

Part X — Applied Physics · Chapter 56

Energy Systems

How physical principles govern the production, conversion, storage, transmission, and use of energy in modern civilisation

Every light bulb, every electric vehicle, every heated home, every manufactured product, and every data centre draws on an energy system: a chain of physical processes that converts primary energy in nature into useful work and heat at the point of use. Coal burns to heat steam; steam expands through a turbine; the turbine drives a generator; the generator induces an EMF that drives current through transmission lines; transformers step voltages up and down; and finally the energy arrives at a socket where it can do work. At every step, the laws of thermodynamics, electromagnetism, fluid mechanics, and nuclear physics govern what is possible, how much is lost, and what the environmental consequences are. This chapter is a synthesis of the entire course, showing how the physics you have studied combines in the infrastructure that powers civilisation.

56.1 What an Energy System Is

An energy system is the complete chain of processes that extracts energy from nature, converts it into useful forms, stores it where needed, transmits it to where it is needed, and delivers it for final use. Energy systems involve primary energy sources (the forms in which energy exists in nature), energy carriers (intermediate forms used to transport or store energy), and end-use technologies (devices that deliver the services users actually want: heat, light, mechanical work, communication).

The key physical constraint on all energy systems is the laws of thermodynamics. The first law (energy conservation) requires that energy input equals useful output plus all losses. The second law requires that some energy is always degraded to low-grade heat in any real conversion process; the Carnot efficiency sets the upper bound for thermal machines. No energy system delivers 100% useful output; real systems have cascading losses at every conversion step.

Energy is measured in joules (J), but practical energy systems use larger units. 1 kWh (kilowatt-hour) = 3.6 MJ. 1 toe (tonne of oil equivalent) = 41.87 GJ. 1 quad = 10^{15} BTU $\approx$ 1.055 EJ (used in US energy statistics). Global primary energy consumption is approximately 580 EJ/year (as of the mid-2020s), equivalent to a continuous global power use of about 18 TW.

Unit	Equivalent in Joules	Typical Context
1 calorie (nutritional)	4186 J = 4.186 kJ	Food energy
1 kWh	3.6×10^6 J = 3.6 MJ	Household electricity billing
1 toe (tonne oil equivalent)	4.187×10^{10} J = 41.87 GJ	Oil and gas statistics
1 EJ (exajoule)	10^{18} J	National/global energy statistics
1 TWh (terawatt- hour)	3.6×10^{15} J = 3.6 PJ	Grid-scale electricity
1 Mt TNT equivalent	4.184×10^{15} J	Nuclear weapon yield comparison

Table 56.1 *Energy unit conversions used in energy systems analysis.*

56.2 Primary vs Secondary Energy Sources

Primary energy sources are found in nature before any human conversion. They include fossil fuels (coal, oil, natural gas — stored solar energy from ancient photosynthesis), nuclear fuels (uranium, thorium — nuclear binding energy), and renewable flows (solar radiation, wind, falling water, geothermal heat, tidal energy). Primary energy is characterised by energy density (joules per kilogram or joules per litre) and by whether it is a stock (finite quantity that is depleted) or a flow (continuously replenished).

Secondary energy carriers are derived from primary sources through conversion: electricity (from generators driven by thermal, hydraulic, wind, or nuclear power), hydrogen (from electrolysis or steam methane reforming), refined petroleum products (from crude oil), and district heat (from co-generation plants). Secondary carriers are more useful than primary sources — they are portable, controllable, and deliverable through infrastructure networks — but conversion always involves losses.

Source	Type	Energy Density	CO ₂ Intensity (gCO ₂ /kWh)	Key Physics
Coal (hard)	Fossil stock	≈ 29 MJ/kg	820–900	Combustion, Rankine cycle
Natural gas	Fossil stock	≈ 55 MJ/kg	490–550	Combustion, CCGT turbines
Crude oil	Fossil stock	≈ 44 MJ/kg	650–750	Combustion, Otto/diesel cycle
Uranium (in LWR)	Nuclear stock	≈ 45 GJ/kg	≤ 15	Fission, Rankine cycle
Wind	Renewable flow	Variable (0.5 kW/m ² typical)	≤ 10	Bernoulli, Betz limit
Solar PV	Renewable flow	≈ 170 W/m ² peak	≤ 30	Photovoltaic effect
Hydro	Renewable flow	Site-dependent	≤ 15	Potential energy, Pelton/Kaplan
Geothermal	Renewable flow	Site-dependent	15–100	Heat engines (low ΔT)

Table 56.2 *Primary energy sources with key physical and environmental characteristics.*

56.3 Mechanical Energy Systems

The simplest energy systems convert gravitational or kinetic energy to mechanical work directly. Waterwheels, windmills, and sailing ships are ancient examples; their modern descendants are hydroturbines, wind turbines, and wave energy converters.

Mechanical energy systems are governed by conservation of energy: $E_k + E_p = \text{constant}$ (no losses), with real systems losing energy to friction, turbulence, and material vibration. The power available from a mechanical energy flow is $P = Fv$ (linear motion) or $P = \tau\omega$ (rotational motion), where τ is torque and ω is angular velocity.

Flywheels store rotational kinetic energy: $E = \frac{1}{2} I\omega^2$, where I is the moment of inertia. Modern flywheel energy storage systems use high-strength composite rotors spinning at 20 000–60 000 rpm in vacuum enclosures, storing tens of kWh with round-trip efficiencies of 90–95%. They are used for frequency regulation in electrical grids and for uninterruptible power supplies.

Example — Flywheel energy storage A flywheel has $I = 50 \text{ kg}\cdot\text{m}^2$ and spins at 30 000 rpm = 3142 rad/s. Energy stored: $E = \frac{1}{2} \times 50 \times 3142^2 = \frac{1}{2} \times 50 \times 9.87 \times 10^6 = 2.47 \times 10^8 \text{ J} = 247 \text{ MJ} \approx 68.5 \text{ kWh}$. This is enough to power an average home for about 2–3 days.

56.4 Thermal Power Generation

The majority of the world's electricity is generated by thermal power plants, which convert heat from combustion (fossil fuels, biomass) or nuclear fission into electrical energy via a heat engine driving a generator. The working fluid in most thermal power plants is steam (water vapour), cycled through the Rankine cycle: heating at constant pressure, expansion through a turbine (work output), condensation back to liquid, and pumping back to high pressure.

$$\eta_{\text{Carnot}} = 1 - T_{\text{C}} / T_{\text{H}}$$

Carnot efficiency: the maximum possible efficiency of any heat engine operating between hot reservoir temperature T_{H} and cold reservoir temperature T_{C} (both in kelvin). Real power plants achieve 30–45% of this limit due to irreversibilities. T_{H} limited by material strength (steam turbines: ≈ 60 MPa, 600°C); T_{C} limited by cooling water temperature.

A modern supercritical coal power plant operates at $T_{\text{H}} \approx 873$ K (600°C) and $T_{\text{C}} \approx 308$ K (35°C). Carnot efficiency: $1 - 308/873 = 64.7\%$. Actual efficiency: $\approx 43\text{--}46\%$ (due to friction, heat losses, pump and turbine irreversibilities). A combined cycle gas turbine (CCGT) plant uses a gas turbine (Brayton cycle) at high temperature and recovers waste heat in a steam turbine (Rankine cycle), achieving efficiencies of 55–60%.

All thermal power plants must reject waste heat to a cold reservoir — typically a river, lake, ocean, or cooling tower. This waste heat is not a design failure but a thermodynamic necessity. A 1 GW electrical plant at 40% efficiency must reject 1.5 GW of waste heat (since $P_{\text{in}} = P_{\text{out}} + P_{\text{waste}}$; $P_{\text{waste}}/P_{\text{out}} = (1-\eta)/\eta = 1.5$ at 40%). Cooling towers are concrete hyperbolic structures that transfer this heat to the atmosphere by evaporation. The white plume from a cooling tower is water vapour, not pollution.

Example — Thermal plant efficiency A coal plant burns 1 kg of coal (heating value 29 MJ/kg) per second. Electrical output = $29 \text{ MW} \times 0.40$ (efficiency) = 11.6 MW. Waste heat rejected = $29 - 11.6 = 17.4$ MW. CO_2 emission: coal is approximately 85% carbon; 0.85 kg C per second produces $0.85 \times 44/12 = 3.12$ kg $\text{CO}_2/\text{s} = 3.12/(11.6 \times 10^{-3}) = 269$ g CO_2/kWh .

56.5 Hydroelectric and Fluid Power Systems

Hydroelectric power converts the gravitational potential energy of elevated water into electrical energy. Water falling through height h releases potential energy $E = mgh$ per unit mass. A turbine converts this kinetic energy of the water flow into rotational mechanical energy, which drives a generator.

$$P = \eta \rho g Q h$$

Hydroelectric power. P = electrical output (W); η = overall efficiency (≈ 0.85 – 0.92 for modern plants); ρ = water density (1000 kg/m^3); $g = 9.81 \text{ m/s}^2$; Q = volume flow rate (m^3/s); h = hydraulic head (m, effective height of fall).

Hydroelectric plants range from small run-of-river systems (no significant reservoir, head 2–10 m, power tens of kW) to large impoundment facilities (reservoir stores months of flow, head hundreds of metres, power gigawatts). The Three Gorges Dam in China (22.5 GW capacity) is the world's largest hydroelectric facility, generating about 100 TWh/year — enough to supply roughly 15 million average Chinese households.

Pumped-storage hydroelectricity (PSH) is the dominant form of large-scale electrical energy storage worldwide. During periods of excess generation (low demand, high renewable output), electricity pumps water uphill to a reservoir. During peak demand, water is released to generate electricity. Round-trip efficiency is approximately 75–85%. PSH accounts for about 94% of the world's installed grid energy storage capacity.

Tidal energy exploits the potential and kinetic energy of ocean tides, which arise from the gravitational forces of the Moon and Sun on the oceans. Tidal barrages (La Rance, France, 240 MW) capture tidal flow through sluice gates and turbines. Tidal stream generators (analogous to underwater wind turbines) extract kinetic energy from tidal currents where flow is fast (e.g., the Pentland Firth, UK, with currents up to 4 m/s).

Example — Hydroelectric power A river flows at $Q = 250 \text{ m}^3/\text{s}$ through a turbine with hydraulic head $h = 80 \text{ m}$. Efficiency = 0.88. Power = $0.88 \times 1000 \times 9.81 \times 250 \times 80 = 172.7 \times 10^6 \text{ W} \approx 173 \text{ MW}$. This is a medium-sized hydroelectric plant.

56.6 Solar Energy Systems

Solar energy reaches Earth's surface at a mean irradiance of about 170 W/m^2 globally (averaged over day/night, latitude, and weather). Peak irradiance on a horizontal surface can reach 1000 W/m^2 under full sun in the tropics. Two main technologies convert solar radiation to useful energy: photovoltaics (PV, converting light directly to electricity) and solar thermal (concentrating heat for hot water, space heating, or electricity generation).

Photovoltaic cells use the photoelectric effect in semiconductor p-n junctions. Incident photons with energy $E = hf \geq E_{\text{gap}}$ excite electrons across the band gap, creating electron-hole pairs that are separated by the built-in electric field of the junction and collected as direct current. Silicon single-crystal cells have band gaps of 1.12 eV ; this matches the solar spectrum well, but photons with energy less than 1.12 eV (infrared) are not absorbed, and photons with energy much greater than 1.12 eV (UV) deliver most of their energy as waste heat. These losses account for the theoretical Shockley-Queisser efficiency limit of about 33% for a single-junction silicon cell.

$$E_{\text{photon}} = hf = hc / \lambda \quad (\text{minimum for PV absorption})$$

$$E_{\text{photon}} \geq E_{\text{gap}}$$

Photovoltaic energy condition. $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$; $c = 3 \times 10^8 \text{ m/s}$; $f = \text{frequency (Hz)}$; $\lambda = \text{wavelength (m)}$. For silicon ($E_{\text{gap}} = 1.12 \text{ eV} = 1.79 \times 10^{-19} \text{ J}$), the threshold wavelength is $\lambda_{\text{max}} = hc/E_{\text{gap}} \approx 1110 \text{ nm}$ (near-infrared). Only photons with $\lambda < 1110 \text{ nm}$ can excite electrons.

Commercial silicon PV cells achieve efficiencies of 20–23%. Multi-junction concentrator cells (used in spacecraft and concentrated solar power) achieve over 40%. The capacity factor of solar PV (fraction of time at peak output) is 10–25% in most locations, determined by solar resource and local weather. The cost of solar PV has fallen by over 90% since 2010, making it the cheapest source of new electricity generation in much of the world as of the mid-2020s.

Concentrated solar power (CSP) uses mirrors or lenses to concentrate sunlight onto a receiver, generating high-temperature heat that drives a conventional steam turbine. CSP plants can incorporate thermal storage (molten salt at $390\text{--}560^\circ\text{C}$) that allows electricity generation for hours after sunset. Parabolic trough, power tower (heliostat field), and linear Fresnel designs are the main CSP configurations.

56.7 Wind Energy Systems

Wind turbines extract kinetic energy from moving air. The power available in a wind stream passing through area A at speed v is $P_{\text{wind}} = \frac{1}{2}\rho A v^3$ (kinetic energy flux). No turbine can extract all of this: the Betz limit (derived from conservation of momentum applied to the actuator disc model) states that the maximum extractable fraction is $16/27 \approx 59.3\%$.

$$P_{\text{extracted}} = \frac{1}{2} \rho A v^3 \times C_p \quad (C_p \leq 16/27 \approx 0.593)$$

Wind turbine power: ρ = air density ($\approx 1.2 \text{ kg/m}^3$); A = rotor swept area = πR^2 (m^2); v = wind speed (m/s); C_p = power coefficient (dimensionless, ≤ 0.593 by Betz limit). Modern large turbines achieve $C_p \approx 0.45\text{--}0.50$. Note the cubic dependence on wind speed: doubling wind speed increases available power eightfold.

Modern onshore wind turbines have rotor diameters of 100–175 m and rated capacities of 3–6 MW. Offshore wind turbines reach 15–22 MW (rotor diameters up to 280 m). Capacity factors for onshore wind are typically 25–40%; offshore 40–60%. Wind turbines operate between cut-in speed ($\approx 3 \text{ m/s}$, below which the blades do not generate useful power) and cut-out speed ($\approx 25 \text{ m/s}$, above which the turbine is shut down to prevent damage).

Example — Wind turbine power A turbine with rotor diameter 120 m operates in 10 m/s wind. Swept area $A = \pi \times 60^2 = 11\,310 \text{ m}^2$. Wind power available = $\frac{1}{2} \times 1.2 \times 11\,310 \times 10^3 = 6.786 \times 10^6 \text{ W} = 6.79 \text{ MW}$. At $C_p = 0.48$: output = $6.79 \times 0.48 = 3.26 \text{ MW}$. At 15 m/s (50% more wind): power $\propto v^3$, so output = $3.26 \times (15/10)^3 = 3.26 \times 3.375 = 11.0 \text{ MW}$.

The tip-speed ratio (TSR) $\lambda = \omega R/v$ is the ratio of blade tip speed to wind speed. Optimal TSR for maximum C_p is about 6–8 for modern three-bladed turbines. At TSR < optimal, the rotor turns too slowly to sweep all the wind efficiently; at TSR > optimal, turbulence from one blade interferes with the next.

56.8 Nuclear Energy Systems

Nuclear fission releases energy when heavy nuclei (uranium-235, plutonium-239) absorb a neutron and split into two smaller nuclei plus 2–3 fast neutrons and energy. The energy per fission event is about $200 \text{ MeV} = 3.2 \times 10^{-11} \text{ J}$. This is about 50 million times more energy per reaction than chemical combustion; 1 kg of U-235 undergoing complete fission releases the equivalent energy of about 2700 tonnes of coal.

$$E = \Delta m c^2$$

Mass-energy equivalence. Δm = mass defect (the difference in mass between reactants and products); $c = 3 \times 10^8 \text{ m/s}$. For U-235 fission, $\Delta m/m \approx 9 \times 10^{-4}$ (0.09% of mass converted to energy). This small fraction yields enormous energy due to $c^2 = 9 \times 10^{16} \text{ m}^2/\text{s}^2$.

In a nuclear reactor, a controlled chain reaction is maintained. Each fission event produces 2–3 neutrons; exactly one must on average cause another fission ($k = 1$, criticality) for a sustained reaction. Moderators (water, heavy water, graphite) slow fast neutrons to thermal energies where U-235 fission cross-section is much higher. Control rods (boron, hafnium, cadmium) absorb neutrons and allow the reaction rate to be regulated. The heat produced heats coolant (water, heavy water, sodium, helium), which drives a steam turbine through a Rankine cycle.

Nuclear power plants produce no CO_2 during operation. Life-cycle emissions (including construction, fuel mining, and decommissioning) are 5–15 gCO_2/kWh , comparable to wind and lower than solar PV. The main challenges are: high capital cost and long construction time, management of spent fuel (high-level radioactive waste with long-lived isotopes), and public concern about safety (following accidents at Three Mile Island 1979, Chernobyl 1986, and Fukushima 2011). Nuclear fusion (Chapter 53) remains a long-term research goal.

Reactor Type	Moderator	Coolant	Key Characteristics
Pressurised Water Reactor (PWR)	Light water	Light water (pressurised)	Most common globally; ~70% of fleet
Boiling Water Reactor (BWR)	Light water	Light water (boils in core)	Simpler; steam goes directly to turbine
CANDU (PHWR)	Heavy water (D_2O)	Heavy water	Uses natural (unenriched) uranium
High Temperature Gas Reactor	Graphite	Helium	Higher efficiency; inherently safe
Fast Breeder	None (fast)	Liquid sodium	Breeds Pu-239 from U-

Reactor	neutrons)		238; uses more fuel
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Table 56.3 *Main nuclear reactor types and their key physical characteristics.*

56.9 Batteries and Energy Storage

Energy storage is essential for balancing supply and demand in electrical grids (especially with variable renewables), for mobile applications (electric vehicles), and for backup power. Batteries convert chemical energy to electrical energy through electrochemical reactions at two electrodes separated by an electrolyte.

The voltage of a battery cell is determined by the electrochemical potential difference between the anode and cathode materials. Energy density (Wh/kg or Wh/L) determines how much energy can be stored per unit mass or volume. Power density (W/kg) determines how fast energy can be delivered. Cycle life is the number of charge-discharge cycles before capacity degrades significantly.

Battery Type	Voltage (V)	Energy Density (Wh/kg)	Cycle Life	Applications
Lead-acid	2.0/cell	30–50	200–500	Car starter, UPS, low-cost storage
Nickel-metal hydride (NiMH)	1.2/cell	60–90	500–1000	Hybrid vehicles (Prius), consumer electronics
Lithium-ion (LFP)	3.2/cell	100–160	2000–6000	Grid storage, EVs (long-range focus)
Lithium-ion (NMC)	3.7/cell	150–220	1000–2000	EVs (energy density focus), portable electronics
Lithium-ion (NCA)	3.6/cell	200–260	500–1500	High-performance EVs (Tesla 18650)
Sodium-ion	3.1–3.3/cel	100–150	1000–3000	Stationary

	I			storage, lower cost alternative
Flow battery (vanadium)	Variable	15–30 (system)	>10 000	Grid-scale, long duration storage

Table 56.4 *Battery technologies and their key characteristics for energy storage applications.*

The specific energy of a battery cell is limited by the electrochemistry: $E = Q \times V$, where Q is the charge capacity (coulombs) and V is the cell voltage. For lithium-ion, the theoretical specific energy of the active materials alone is about 400–600 Wh/kg; actual cells deliver 150–260 Wh/kg because they include inactive mass (current collectors, casing, electrolyte, separators). Improving specific energy requires either finding electrode materials with higher capacity (Ah/kg) or higher voltage, or reducing inactive mass.

56.10 Electrical Transmission and Distribution

Electrical power is transmitted from generating stations to consumers through a network of transmission lines (high voltage, long distance) and distribution lines (lower voltage, local). Power loss in a transmission line is $P_{\text{loss}} = I^2 R$, where R is the line resistance and I is the current. For fixed power $P = IV$, increasing voltage reduces current in proportion, reducing losses by I^2 .

$$P_{\text{loss}} = I^2 R = P^2 R / V^2$$

Transmission line power loss. P = transmitted power (W); I = current (A); R = line resistance (Ω); V = transmission voltage (V). Doubling the transmission voltage reduces losses by a factor of four for the same power level. High-voltage AC (132–765 kV) and HVDC (400–800 kV) minimise losses over long distances.

Transformers (Chapter 48) step voltage up at the generating station (e.g., from 11 kV to 400 kV for long-distance transmission) and step it down progressively for distribution and final use. A typical UK domestic supply is 230 V (single-phase); US is 120/240 V. Industrial supplies are 11 kV or 33 kV. The efficiency of modern power transformers is very high (99.5–99.7%), making them one of the most efficient energy conversion devices.

Example — Transmission line losses A power station generates 500 MW and transmits it through a 400 kV, 300 km line with resistance 0.06 Ω /km (total $R = 18 \Omega$). Current $I = P/V = 500 \times 10^6 / 400 \times 10^3 = 1250$ A. Power loss = $I^2 R = 1250^2 \times 18 = 28.1$ MW = 5.6% of transmitted power. If transmitted at 11 kV (36 times lower): $I = 45\,455$ A; loss = $45455^2 \times 18 = 37.2$ GW — vastly exceeding the generated power, which

shows why high voltage transmission is essential.

High-voltage direct current (HVDC) transmission uses power electronics (rectifiers and inverters) to convert AC to DC for long-distance transmission and back to AC for distribution. HVDC has lower line losses than HVAC for distances over ≈ 600 km (overhead) or ≈ 100 km (underground/submarine cable), because capacitive charging losses in AC cables are eliminated. HVDC is preferred for long submarine cables (e.g., interconnectors between countries) and for integrating distant renewable resources.

56.11 Efficiency and Energy Losses in Infrastructure

Every stage of an energy system loses some energy, typically as heat. The overall efficiency of a system is the product of the efficiencies of each stage. A typical electricity-to-heat chain in an old electric resistance heater: power plant (40%) $\times$ transmission (95%) $\times$ distribution (97%) $\times$ heater (100%) = 36.9% primary energy to useful heat. Replacing the resistance heater with a heat pump (COP 3.0) gives: 40% $\times$ 95% $\times$ 97% $\times$ 300% = 110.7% — more useful heat than the primary energy input to the power plant, because the heat pump also draws heat from the environment.

$$\eta_{\text{system}} = \eta_1 \times \eta_2 \times \eta_3 \times \dots$$

System efficiency is the product of all stage efficiencies. Each stage multiplies the fraction of energy delivered. Even if each stage is individually efficient, cascading losses can make the overall system significantly inefficient.

The coefficient of performance (COP) of a heat pump is the ratio of useful heat output to electrical work input: $\text{COP} = Q_H / W$. For a Carnot heat pump: $\text{COP}_{\text{max}} = T_H / (T_H - T_C)$. A ground-source heat pump with $T_C = 7^\circ\text{C}$ (280 K) and $T_H = 35^\circ\text{C}$ (308 K) has $\text{COP}_{\text{max}} = 308/28 = 11.0$; practical COPs are 3–5.

Electric vehicles (EVs) are substantially more energy-efficient than internal combustion engine (ICE) vehicles. An ICE vehicle converts about 20–25% of fuel energy to wheel motion (the rest is heat). An EV converts about 85–90% of battery energy to wheel motion. Even accounting for power-plant losses in generating the electricity, EVs typically deliver 2–3 times more useful mobility per unit of primary energy than ICE vehicles, and 4–6 times more when powered by renewable electricity.

Energy System	Primary-to-Service Efficiency	Notes
Incandescent bulb (electricity from coal)	$\approx 4\%$	Plant 40% $\times$ grid 95% $\times$ bulb 10%

LED bulb (electricity from coal)	$\approx 36\%$	Plant 40% $\times$ grid 95% $\times$ LED 95%
LED bulb (solar PV, direct)	$\approx 18\%$	PV 20% $\times$ LED 95%
ICE vehicle (petrol)	15–25%	Tank-to-wheel; city driving lower
EV (grid electricity from coal)	$\approx 36\%$	Plant 40% $\times$ grid 95% $\times$ EV 90%
EV (solar PV)	$\approx 17\%$	PV 20% $\times$ EV 90%
Natural gas condensing boiler (space heat)	$\approx 93\%$	Latent heat recovery
Heat pump (ground-source, from coal plant)	$\approx 120\text{--}150\%$	COP 3–5 $\times$ plant/grid efficiency

Table 56.5 *Approximate primary-to-service efficiencies for common energy chains.*

56.12 Grid Stability and Energy Demand

An electrical grid must balance supply and demand instantaneously — electricity cannot easily be stored in large quantities, and if supply exceeds demand, frequency

risers; if demand exceeds supply, frequency falls. Grid frequency (50 Hz in Europe, 60 Hz in North America) is the real-time indicator of this balance. Generators slow down if they are overloaded (demand > supply); they speed up if underloaded (supply > demand).

Grid stability is maintained by a hierarchy of responses: primary frequency response (automatic, within seconds, from generator governor systems); secondary frequency control (automatic generation control, minutes timescale); and tertiary control (manual dispatch of reserve generation, tens of minutes). Large inertia (from spinning generator masses) provides the system with time to respond before frequency deviates dangerously. As fossil-fuel synchronous generators are replaced by variable-speed inverter-based renewable generators, traditional rotational inertia is reduced — requiring new approaches such as virtual inertia from grid-scale batteries.

Demand variability follows predictable daily patterns (morning and evening peaks), seasonal patterns (winter heating demand, summer cooling demand), and unpredictable short-term fluctuations (e.g., large simultaneous kettle use at the end of a popular TV show — a real phenomenon that UK grid operators must plan for). Demand response programmes incentivise large electricity consumers to shift or curtail usage during peak demand periods.

Integrating large amounts of variable renewable generation (wind and solar) increases the challenge of balancing because their output depends on weather. Solutions include: flexible backup generation (gas peakers, hydro), demand response, energy storage (batteries, pumped hydro, hydrogen), and interconnection with neighbouring grids so surplus renewable generation in one region can serve demand in another. Long-duration energy storage (8–100 hours) is an active area of technology development.

Key Point: The term “baseload” — generation that runs continuously at steady output — becomes less meaningful in a grid with high renewable penetration. What matters is flexibility: the ability to ramp generation up and down quickly in response to changes in renewable output and demand. Physics determines the ramp rates: hydro and gas can ramp in seconds to minutes; coal plants take tens of minutes to hours.

56.13 Common Mistakes in Energy Systems Analysis

- Confusing power and energy. Power (watts, W) is the rate of energy flow; energy (joules, J, or kilowatt-hours, kWh) is the total amount. A 3 GW nuclear plant produces 3 GW of power; in one year it produces $3 \text{ GW} \times 8760 \text{ h} \times 0.90$

(capacity factor) = 23 652 GWh = 23.7 TWh of energy. Capacity and annual generation are different quantities.

- Applying efficiency to the wrong quantity. Carnot efficiency limits the fraction of heat that can be converted to work; it does not apply to energy flows that do not involve heat-to-work conversion (e.g., PV panels, fuel cells, batteries). Applying the Carnot limit to a battery is a category error.
- Forgetting the v^3 dependence of wind power. A 10% increase in average wind speed increases available wind power by $(1.10)^3 \approx 1.33$ — a 33% increase in power, not 10%. Wind resource assessment errors compound rapidly.
- Thinking renewable generation capacity equals renewable generation. A 1 GW solar farm with a 15% capacity factor generates $1 \text{ GW} \times 8760 \text{ h} \times 0.15 = 1314 \text{ GWh/year}$. A 1 GW nuclear plant with 90% capacity factor generates 7884 GWh/year — six times more energy, from the same installed capacity.
- Confusing CO₂ per kWh (generation emissions intensity) with total CO₂ per year (total emissions). Total emissions depend on both intensity and how much energy is generated. A very low-emissions source (nuclear) that generates a lot can have a larger impact on decarbonisation than a zero-emissions source (offshore wind) that generates less in a specific context.
- Assuming 100% renewable grids are straightforward. Matching variable supply to demand at all hours requires storage, interconnection, flexible demand, or dispatchable backup. The physics of energy storage (battery energy density, pumped hydro geography) and transmission (line capacity, losses) constrain what is achievable and at what cost.

Common Mistake: Never use efficiency without specifying the input and output energy forms. Thermal efficiency, electrical efficiency, exergy efficiency, and round-trip efficiency all measure different things and can give very different numbers for the same system. Always state: efficiency of converting what, to what, under what conditions.

Chapter Summary

Energy systems chain together primary sources, conversion technologies, transmission infrastructure, storage, and end-use devices. At every stage, thermodynamics, electromagnetism, fluid mechanics, and nuclear physics determine efficiency and loss.

- Energy and power: E (joules, kWh) vs P (watts); system efficiency = product of all stage efficiencies; second law limits all thermal conversions.
 - Thermal power: Rankine cycle; Carnot limit $\eta = 1 - T_C/T_H$; CCGT achieves 55–60%; all plants reject waste heat to cold reservoir.
 - Hydro: $P = \eta \rho g Q h$; pumped storage for grid-scale electricity storage; tidal barrages and tidal stream.
 - Solar PV: photovoltaic effect; threshold wavelength from E_{gap} ; Shockley-Queisser limit $\approx 33\%$; cost has fallen 90%+ since 2010.
 - Wind: $P = \frac{1}{2} \rho A v^3 \times C_p$; Betz limit $C_p \leq 16/27$; cubic wind speed dependence; capacity factor 25–60%.
 - Nuclear: $E = \Delta mc^2$; 200 MeV per fission; 50 million times more energy-dense than coal; controlled by $k = 1$ criticality; low CO₂ lifecycle.
 - Batteries: electrochemical energy storage; energy density limited by electrode chemistry; lithium-ion dominant; flow batteries for long-duration grid storage.
 - Transmission: $P_{\text{loss}} = I^2 R = P^2 R / V^2$; high voltage reduces losses; transformers step voltage; HVDC for long submarine cables.
 - Grid stability: frequency as real-time supply-demand indicator; inertia buffers response time; variable renewables reduce inertia; storage and flexibility needed.
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Key Terms

primary energy: Energy in its natural form before human conversion: fossil fuels, nuclear fuels, and renewable flows (solar, wind, hydro, geothermal, tidal).

energy carrier: An intermediate form of energy derived from primary sources and used to transport or store energy: electricity, hydrogen, heat, petroleum products.

Rankine cycle: The thermodynamic cycle of a steam power plant: heat input at constant pressure, adiabatic expansion through a turbine, condensation, and pumping back to high pressure.

Carnot efficiency: $\eta_{\text{Carnot}} = 1 - T_{\text{C}}/T_{\text{H}}$; the maximum possible efficiency of a heat engine; depends only on the temperatures of the hot and cold reservoirs; always less than 1.

hydraulic head (h): The effective vertical height through which water falls in a hydroelectric system; determines the potential energy per unit mass (gh) available for conversion.

photovoltaic effect: The generation of electric current when photons with energy $\geq E_{\text{gap}}$ excite electrons across a semiconductor band gap, producing electron-hole pairs that are separated by a p-n junction.

Shockley-Queisser limit: The theoretical maximum efficiency of a single-junction solar cell ($\approx 33\%$ for silicon); set by the band-gap mismatch with the solar spectrum.

Betz limit: $16/27 \approx 59.3\%$; the maximum fraction of wind kinetic energy that any wind turbine can extract; derived from conservation of momentum applied to the rotor disc.

power coefficient (C_p): The ratio of extracted wind power to available wind power; $C_p \leq 16/27$ (Betz limit); modern turbines achieve $C_p \approx 0.45\text{--}0.50$.

mass defect (Δm): The difference in mass between reactants and products in a nuclear reaction; the lost mass appears as energy $E = \Delta mc^2$; positive for exothermic reactions (fission, fusion).

criticality ($k = 1$): The condition in a nuclear reactor where exactly one neutron from each fission event causes another fission, sustaining a steady chain reaction.

energy density: Energy stored or released per unit mass (Wh/kg, MJ/kg) or per unit volume (Wh/L, MJ/L); determines the mass and volume of fuel or battery required for a given energy storage.

capacity factor: The ratio of actual energy generated over a period to the maximum possible energy if the plant ran at full rated capacity continuously; accounts for downtime, variable resource, and part-load operation.

grid frequency: The frequency of alternating current on an electrical grid (50 Hz in Europe, 60 Hz in North America); maintained at the reference value by continuous matching of generation to demand.

coefficient of performance (COP): For a heat pump: $\text{COP} = Q_{\text{H}}/W$ (heat delivered divided by electrical work input); can exceed 1 because heat is moved from a cold source, not created from scratch.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. A 1 GW coal power plant has a thermal efficiency of 40% and a 90% capacity factor. How much coal (heating value 29 MJ/kg) does it consume per year? How much waste heat does it reject per year?
2. Explain why high-voltage transmission (400 kV) reduces power losses compared to lower-voltage transmission (11 kV) for the same power level. Use the formula $P_{\text{loss}} = I^2R$ to support your explanation.

3. Explain the Betz limit. Why can't a wind turbine extract all the kinetic energy from the wind passing through it? What would happen to the air behind the turbine if all kinetic energy were extracted?
 4. Compare the energy density of uranium in a light water reactor to coal (coal: 29 MJ/kg; U-235 complete fission: 80 TJ/kg). By what factor is nuclear more energy-dense? What practical implication does this have?
 5. A pumped-storage hydroelectric plant stores energy by pumping 10 million cubic metres of water uphill by 400 m. How much energy does it store in GWh? (Round-trip efficiency = 80%.) How long could this supply a 1 GW demand?
 6. Explain the difference between a battery's energy density (Wh/kg) and power density (W/kg). Why might a battery with high energy density but low power density be unsuitable for a vehicle acceleration application?
 7. Why does increasing variable renewable penetration on an electrical grid create challenges for frequency stability, even if total annual generation is sufficient? What physical quantity, previously provided by fossil-fuel generators, is reduced?
 8. A household installs a ground-source heat pump with COP = 4.0 instead of an electric resistance heater. The electricity is generated at a 40% efficient gas power plant. Compare the primary gas energy consumption per MJ of heat delivered for each option.
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Power Plant Analysis

9. A coal power plant (efficiency 38%) burns coal with heating value 28.5 MJ/kg. It generates 850 MW of electricity. Find: (a) the primary thermal power input; (b) the mass of coal burned per second; (c) the waste heat rejected per second; (d) the CO₂ emission rate (coal is 80% carbon by mass).
10. A CCGT plant operates with a gas turbine at $T_H = 1400^\circ\text{C}$ and exhaust at 550°C (this heat goes to a steam turbine with cold reservoir at 30°C). Find the Carnot efficiency of each stage and estimate the maximum overall efficiency of the combined cycle.

11. A nuclear power station (PWR) uses fuel with an effective energy release of 45 GJ per kg of uranium. It operates at 33% thermal efficiency with capacity factor 90% and rated electrical output 1.2 GW. Find: (a) annual electrical generation (TWh); (b) annual thermal energy input (PJ); (c) annual uranium fuel consumption (tonnes).
12. A hydro plant has a head of 120 m, flow rate of 400 m³/s, and efficiency 91%. (a) Find the electrical output in MW. (b) How many tonnes of CO₂ per year would an equivalent coal plant (38% efficiency, 0.9 kg CO₂/kWh electrical) emit?
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Efficiency Problems

13. A household uses 3500 kWh/year for heating. Compare the annual primary energy input (coal at power plant, 40% efficiency; grid transmission 95%; distribution 97%) for: (a) electric resistance heaters; (b) a heat pump with COP = 3.5.
14. An electric vehicle has battery capacity 75 kWh, drivetrain efficiency 88%, and energy consumption 15 kWh/100 km. The charging efficiency is 92%. If electricity is generated from natural gas at 50% efficiency, find the primary energy (MJ) per

- 100 km of driving. Compare to a 6 L/100 km petrol car (petrol heating value 32 MJ/L, ICE efficiency 22%).
15. LED lighting converts 95% of electrical energy to light; incandescent bulbs convert 10%. If electricity is generated from solar PV at 20% efficiency, find the primary solar energy (W/m^2 of panel) needed to produce 800 lumens of light from each bulb type. (Assume 100 lm/W for LED, 15 lm/W for incandescent, and standard irradiance of 170 W/m^2 annual average.)
16. A steam turbine takes in steam at 500°C (773 K) and exhausts at 40°C (313 K). The Carnot efficiency is 59.5%, but the actual isentropic efficiency of the turbine blades is 88% and mechanical/generator losses are 3%. Find the actual electrical output efficiency of the turbine-generator set.
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Energy Conversion Exercises

17. A wind turbine has rotor diameter 150 m and operates in mean wind speed 9 m/s with $C_p = 0.47$. (a) Find the mean electrical output. (b) Find annual energy generation if capacity factor = 35%. (c) How many households (each using 4500 kWh/year) does this turbine serve?
18. A solar farm covers 200 hectares with PV panels of efficiency 21% and packing factor 0.70 (70% of area covered). Mean irradiance is 180 W/m^2 . Find the mean electrical output and annual generation. (1 hectare = $10\,000 \text{ m}^2$)

19. A uranium-235 fission releases 200 MeV. Find: (a) this energy in joules; (b) the energy per kg of U-235 if complete fission occurs; (c) the mass defect per fission event as a fraction of the U-235 mass ($235 \text{ u} = 235 \times 1.66 \times 10^{-27} \text{ kg}$).
20. A Li-ion battery pack (NMC chemistry) has energy density 200 Wh/kg and powers a vehicle requiring 180 Wh/km. How heavy is the battery needed for a 500 km range? Compare to a petrol tank for the same range at 8 L/100 km (petrol density 0.74 kg/L).
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Grid and Storage Applications

21. A 500 MW power station transmits electricity through a 220 kV, 200 km line (resistance $0.08 \text{ } \Omega/\text{km}$). Find: (a) the current in the line; (b) the power loss; (c) the fraction of power lost; (d) the voltage drop across the line.
22. Grid frequency in the UK is 50 Hz. A sudden loss of a 1.8 GW generator causes frequency to begin falling. The rate of change of frequency (RoCoF) is estimated at 0.6 Hz/s initially. If the system inertia is $H = 3$ seconds (stored kinetic energy /

rated power), find the total synchronous generation connected to the grid. (Use: $\text{RoCoF} = (P_{\text{imbalance}}) / (2H \times P_{\text{system}})$; solve for P_{system} .)

23. A pumped-storage plant has upper reservoir volume $8 \times 10^6 \text{ m}^3$, head 550 m, turbine-generator efficiency 88%, and pump efficiency 87%. (a) Find the gross stored energy (GWh). (b) Find the electrical energy recoverable on discharge (GWh). (c) Find the round-trip efficiency (electricity in to electricity out). (d) If the plant discharges at full power of 1600 MW, how many hours of storage does it provide?
24. A community installs 20 MW of solar PV (capacity factor 13%) and 5 MW of wind (capacity factor 35%) to meet an average demand of 8 MW. (a) Find mean solar and wind generation. (b) Is average generation sufficient? (c) Identify the physical problem and propose two storage or backup solutions, explaining the physics of each.

Chapter 56 Checkpoint

Before moving on to Chapter 57, confirm you can do each of the following.

I can...	See Section
Explain the role of thermodynamics and the second law in all energy systems	56.1–56.2

Calculate Carnot efficiency and actual plant efficiency from operating temperatures	56.4
Apply $P = \eta \rho g Q h$ to hydroelectric power output	56.5
Apply the photovoltaic threshold condition $E_{\text{photon}} \geq E_{\text{gap}}$ to solar cells	56.6
Calculate wind turbine output using $P = \frac{1}{2} \rho A v^3 \times C_p$ and the Betz limit	56.7
Explain nuclear fission energy release using $E = \Delta mc^2$ and the chain reaction	56.8
Compare battery chemistries by energy density, cycle life, and application	56.9
Calculate transmission line losses using $P_{\text{loss}} = I^2 R = P^2 R / V^2$	56.10
Calculate and cascade system efficiencies through multiple conversion stages	56.11
Explain grid frequency as a supply-demand indicator and describe stability challenges	56.12
Identify common mistakes in energy calculations, especially power vs energy	56.13

Continue to Chapter 57: Computational and Data-Based Physics →